

DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE Regular End Semester Examination – Summer 2023 Course: B. Tech. Branch: Mechanical Engineering Semester: IV Subject Code & Name: BTMC401 /Manufacturing Processes – I Max Marks: 60 Date:13/07/2023 Duration: 3.00 Hr.			
Instructions to the Students: 1. All the questions are compulsory. 2. The level of question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in () in front of the question. 3. Use of non-programmable scientific calculators is allowed. 4. Assume suitable data wherever necessary and mention it clearly.			
		(Level/CO)	Marks
Q.1	Solve Any Two of the following.		
A)	Define pattern? State any four types of patterns.	Remember	6
B)	What are the major types of sand moulds? What are their characteristics?	Understand	6
C)	Why is the investment-casting process capable of producing fine surface details on casting?	Understand	6
Q.2	Solve Any Two of the following.		
A)	Define roll gap, neutral point, and draft.	Remember/ Understand	6
B)	What is the difference between cold, warm, and hot forging.	Understand	6
C)	What is forgeability? Explain with neat diagram	Understand	6
Q.3	Solve Any Two of the following.		
A)	What is SMAW? What are the advantages and applications of SMAW?	Understand	6
B)	What is the purpose of flux? Why is it not needed in gas tungsten-arc welding?	Remember	6
C)	Explain the principle involved in MIG welding?	Understand	6
Q.4	Solve Any Two of the following.		
A)	What are the components of a lathe?	Understand	6
B)	What do you mean by tool geometry? How single point turning tool geometry is specified?	Remember	6
C)	Find the time required to drill 4 holes in a cast iron flange each of 2 cm depth, if the	Analysis	6

	hole diameter is 2 cm. Assume cutting speed as 21.9 m/min. and feed as 0.02 cm/rev.		
Q. 5	Solve Any Two of the following.		
A)	Describe the different types of cutters used in milling operations and give an example of each one.	Understand	6
B)	Define broaching. Explain the geometric features of broach with their functions.	Remember	6
C)	Why do machined gears have to be subjected to finishing operations?	Remember	6
	*** End ***		